

Multimodal Embodied AI for Stroke Warning-Sign Assessment

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Abstract—Home-based stroke warning-sign assessment requires responsiveness to abnormal signs with limited disruption during ordinary activity. PiBE-FAST integrates audiovisual perception, guided BE-FAST checks, and rule-based evidence fusion. It preserves reported symptoms across incomplete acquisition, retries, and explicit corrections. Software tests verify these interactions, while participant records characterize target response and nuisance triggering: 35 of 38 scripted targets had recorded triggers, with 0.480 false triggers per reference-negative online hour. These historical records are not linked to current runtime decision paths. The prototype supports evaluation of the monitoring-to-check workflow; reliable stroke warning and reduced final-alert burden remain to be established.

Index Terms—Stroke warning signs, BE-FAST, multimodal sensing, false triggers, edge computing.

I. INTRODUCTION

BE-FAST organizes balance, eyes, face, arms, speech, and symptom-onset information to help recognize possible stroke warning signs [1]. Translating these observations into a home sensing system requires both responsiveness to abnormal signs and restraint during ordinary activity. This work addresses warning signs present during monitoring and guided checks; predicting future stroke incidence is a different task.

Home recordings contain normal movement, changing camera views, background speech, and incomplete actions. A useful warning workflow must distinguish evidence of an abnormal sign from insufficient acquisition. Low positive rates in healthy checks are informative only when considered alongside responses to target abnormalities and the burden of repeated checks. A passive trigger, a guided measurement, and a final user-facing alert therefore require separate evaluation.

PiBE-FAST connects passive Balance and Speech monitoring on a Raspberry Pi with guided B/E/F/A/S checks on a local computer. Speech supplies audio evidence and BEFA supplies video evidence. Personal references characterize within-person change, while quality and action-completion gates determine whether a measurement can be interpreted. Independent symptom reports and onset information contribute to urgency even when sensing fails. Third-party medical-record processing

belongs to the wider project and is outside the evaluated sensing workflow.

Our contribution is an embodied monitoring-to-check prototype that connects AI-based perception with guided interaction and rule-based evidence handling. Three explicit contracts address conflicts in this workflow: incomplete sensing must not erase reported symptoms, retrying a measurement must preserve active symptoms, and correcting a report must retain its history. We evaluate these contracts through state-transition and API tests, and separately characterize recorded target responses, healthy measurements, and deployment behavior. Together, these establish an implemented evaluation platform for the abnormal-sign response–nuisance-triggering trade-off; the effect of guided checks on final alerts requires linked operational outcomes.

II. RELATED WORK

Digital screening systems have combined facial video and speech [2], neurological examination landmarks [3], and multimodal movement evidence [4]. These studies motivate combining observable signs in a screening interface. PiBE-FAST studies their use in a monitoring-to-check workflow, with explicit evaluation of scripted-sign response and reference-negative triggering.

Camera landmarks support contactless measurements of body sway and ocular position [5], [6]. Their interpretation depends on visibility, pose, action completion, and camera conditions. Quiet-standing measures motivate balance assessment [7], whereas clinical ocular assessment requires references beyond uncalibrated RGB geometry [8]. Mandarin corpora support dysarthria modeling [9], and Whisper provides a separate transcription output [10]. PiBE-FAST assigns these measurements distinct roles in acquisition, decision-making, and inspection.

Event sourcing records state changes as events and supports reconstruction of earlier states [11]; provenance models describe the entities, activities, and agents behind data [12]. PiBE-FAST applies this separation of current state and history to BE-FAST-specific contracts: acquisition failure cannot

negate a symptom, ordinary retries cannot retract it, and corrections must preserve earlier reports. The domain specification determines which events change active evidence: a negative retry retains an independent symptom, an explicit correction retracts it, and an urgent report can remain incomplete. These interaction rules support interpretable warning reports when measurements and symptom histories change.

III. SYSTEM OVERVIEW

The embodied AI workflow couples physical sensing, prompted user actions, renewed acquisition, and feedback. The monitoring tier seeks candidate warning signs; guided checks acquire task-specific evidence. The Pi tier continuously observes balance and speech at reduced rates. A retained trigger requests an active check on the local computer, where the user selects or follows B/E/F/A/S tasks. A persistent unresolved-event queue supports delivery retries and acknowledgement. During guided interaction, passive processing is paused as appropriate; the workflow resumes monitoring after the active session.

Figure 1 summarizes the audio/video sensing and feedback workflow. Medical-record processing is implemented externally; its interface and performance are outside this evaluation. The sensing service produces component measurements, acquisition-quality reasons, a research feature vector, and an independent urgency/completeness assessment. Eligible measurements and symptom reports contribute directly to the urgency rule.

Each component record includes status, metrics, quality, and a reason code. Requested tasks can be pending, checking, positive, negative, insufficient, or skipped. A negative result requires completed, qualified acquisition. Technical failure does not itself imply either a clinical abnormality or its absence. Explicit symptom reports carry their own onset metadata. Automatic eye geometry is research-only; the separate visual-symptom questionnaire supplies the eligible E evidence.

The interface presents acquisition progress and the current report, while historical reports retain their original evidence. Photos in the following section illustrate the acquisition setup and live feedback.

IV. MULTIMODAL MEASUREMENTS

Each task applies acquisition-quality and completion gates before interpreting measurements. For sampled tasks, $q_{\text{valid}} = N_{\text{valid}}/N_{\text{captured}}$ uses counts within the component-specific analysis window. Missing landmarks and incomplete actions produce unavailable measurements, with explicit masks in the research representation. The operating points below define the engineering measurement protocol.

A. Balance: Postural Stability

Passive B distinguishes stable standing from ordinary transitions such as sitting, walking, or bending before evaluating change. A separate descent and posture-change path can retain a fall-like trigger. Active B asks the user to stand naturally for 30 s after a 1.5-s warmup. A qualified window requires at least

30 valid samples and valid fraction at least 0.75. Shoulder, hip, and ankle geometry supplies trunk orientation, mediolateral motion, and central sway range. An active balance symptom exempts an unstarted standing attempt. A retry starts a new attempt; a failed earlier attempt remains incomplete in its historical report. Starting a standing task is blocked while a balance symptom remains active, including after an ordinary negative questionnaire answer. Only an explicit correction retracts that event.

Personal enrollment requires a symptom-free attestation and binds the baseline to subject, device, and camera identifiers. The first five qualified windows freeze the median μ_j and scale

$$\sigma_j = \max\{1.4826 \text{ MAD}(X_j), \text{IQR}(X_j)/1.349\}. \quad (1)$$

Changing the enrollment context clears its samples. For trunk orientation θ , lateral-motion velocity v , and central 90% range R , define

$$r_\theta = \frac{|\theta - \mu_\theta|}{\sigma_\theta}, \quad r_v = \frac{\max(0, v - \mu_v)}{\sigma_v}, \quad r_R = \frac{\max(0, R - \mu_R)}{\sigma_R}. \quad (2)$$

The rule is $r_\theta \geq 3.5$ or both $r_v \geq 3.5$ and $r_R \geq 3.5$, using a modified-Z reference [13]. Requiring both sway terms is intended to reduce landmark-jitter triggers. Zero change receives score zero, including at zero scale. Positive change at zero scale is unscorable. A scored positive rule takes precedence; otherwise, an unscorable domain yields insufficient evidence. Figure 3 shows acquisition before the personal baseline has been completed.

B. Eyes: Gaze Assessment

The E task repeats paired center-left and center-right target trials three times. Each target lasts 2 s, with the first 0.5 s reserved for settling. Normalized iris positions produce paired endpoint responses, range, directional asymmetry, and inter-eye differences. Each trial requires five valid samples and valid fraction at least 0.60; each eye must span at least 24 pixels when frame dimensions are available. Repeated responses, head pose, and endpoint dispersion further gate availability. The accompanying measurement protocol fixes these parameters and their definitions. These RGB outputs are research measurements: automatic E disease thresholds are disabled. A separate questionnaire records visual difficulty and its onset, even when camera acquisition fails.

C. Face: Facial Asymmetry

F compares a 2-s neutral phase with a 3-s smile phase, requiring at least five valid samples in each. Mouth-corner displacement is normalized by interocular width; its median change is interpreted alongside left-right smile activation. The trial requires valid fraction at least 0.50 and smile strength at least 0.22. Qualified asymmetry is positive when the absolute neutral-referenced geometric change reaches 0.075 or smile-activation difference reaches 0.24. Failure to establish the action produces insufficient evidence; users or caregivers can separately report inability to smile.

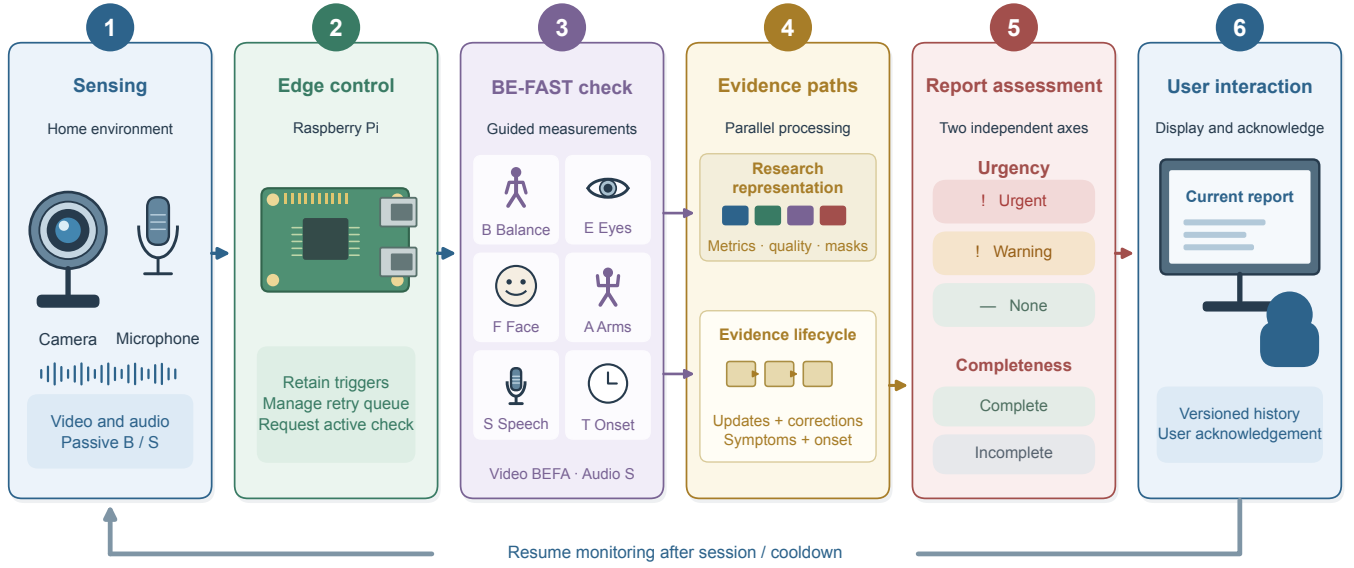


Fig. 1. PiBE-FAST evidence and feedback paths. Research features form a separate branch; urgency and completeness can jointly be urgent and incomplete.

D. Arms: Arm Drift

The A task follows a lower–raise–hold–return sequence, with a 5-s hold near shoulder height (Fig. 2). Bilateral visibility and phase completion are required, with at least 20 visible hold samples and hold-phase valid fraction at least 0.55. The raised-pose gate applies when entering the hold phase. During the hold, all frames with measurable shoulder, elbow, and wrist landmarks are retained, including lowered-arm poses. Early and late segments occupy the first and last thirds of the 5-s interval, with at least six visible samples in each. An unavailable endpoint produces insufficient evidence. With image y increasing downward, shoulder-width-normalized wrist height is $h_{e,t} = (y_{wrist,e,t} - y_{shoulder,e,t}) / w_{shoulder}$. We separate persistent asymmetry from progressive lowering:

$$D = \text{median}_t(h_{L,t} - h_{R,t}), \quad (3)$$

$$\delta_e = \text{median}(h_e^{late}) - \text{median}(h_e^{early}), \quad \Delta = \delta_L - \delta_R. \quad (4)$$

A qualified trial is positive when $|D| \geq 0.30$ or $|\Delta| \geq 0.22$. Anatomical labels assume an unmirrored inference frame; positive differences indicate a lower left arm. Technical inability to obtain this trial remains separate from a reported inability to raise an arm.

E. Speech: Audio Assessment

Passive S compares qualified audio windows with an enrolled personal baseline. The default loop acquires an 8-s window, analyzes it, then waits 1 s; this is sequential window processing. Three abnormal votes are required among the last five qualified windows. From an empty vote history at a window boundary, three 8-s captures and two 1-s waits require at least 26 s plus processing. A target-speaker gate uses three enrollment windows and a cosine-similarity reference of 0.70. The classifier probability contributes only



Fig. 2. Bilateral arm-raise demonstration with live landmark feedback on the local computer.

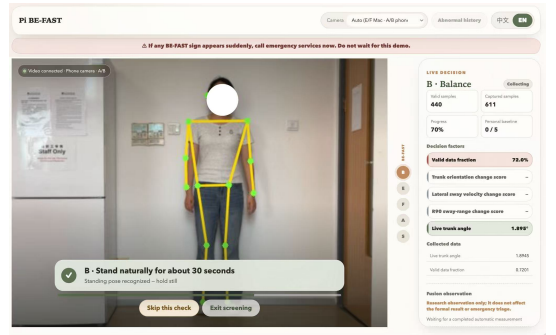


Fig. 3. Balance acquisition with landmark feedback; baseline-relative scores remain unavailable during enrollment (0/5 windows).

one anomaly domain; a trigger additionally requires another changed acoustic domain. The diagnostic output links votes to window identifiers, capture boundaries, and analysis and decision times for subsequent timing audits. The passive gate and the active check share classifier evidence; the active check

adds a new prompted recording and onset information.

Active S acquires a new 7-s, 16-kHz mono recording of a fixed Mandarin sentence. After duration, level, clipping, and voice-content gates, a frozen logistic classifier trained on MDSC supplies dysarthria probability. A value at or above 0.735282 is the direct S positive criterion. Local Whisper transcription supplies character error rate (CER), speaking rate, and pause descriptors for audit; these features do not override the classifier. Dysarthria is not stroke-specific, so measurement onset and independent symptom reports remain necessary inputs to the urgency rule.

V. MULTIMODAL FUSION

Decision-level fusion. The deployed pathway combines eligible component outcomes, independent symptom reports, and onset metadata into urgency and completeness outputs. Passive B/S triggers request guided acquisition; they are distinct from this report assessment. Let M_t contain current measurements and S_t active symptom events. Urgency is *urgent* when any eligible positive has new or unknown onset, *warning* when all positives are explicitly longstanding, and *none* when no eligible positive exists. Completeness independently requires a resolved outcome for every requested component. Thus a new symptom can remain urgent while another task is incomplete. A negative screen requires complete requested evidence without eligible positives.

E supplies questionnaire evidence; camera eye geometry is research-only. B standing exemption applies only to an unstarted task with a reported balance problem. Retries replace measurements without erasing independent symptoms. Corrections require an event identifier and reason, preserving original onset and history; an ordinary negative answer does not retract an earlier symptom. After the last B/E symptom is corrected, its questionnaire awaits a new answer. Asynchronous speech results must match the current recording attempt. Reports retain immutable revisions; configured SQLite history persists positive, reported-functional-problem, and correction reports. These rules are checked separately from sensor accuracy in Section VI.

Research feature fusion. A separate 48-dimensional representation concatenates 19 measurements, five quality values, five component masks, and 19 feature masks:

$$\mathbf{z} = [\mathbf{x}_B, \dots, \mathbf{x}_S, q_B, \dots, q_S, r_B, \dots, r_S, h_1, \dots, h_{19}]. \quad (5)$$

B/E/F/A/S contribute 2/6/3/4/4 measurements. Unavailable features are zeroed with masks set to one, distinguishing missingness from measured zero; original metrics remain in audit records. T supplies onset metadata rather than a sensor feature. This research vector and its severity summaries do not produce the evaluated decisions. The present integration is rule-based evidence fusion, with no demonstrated predictive gain from the research representation.

VI. EVALUATION

We separate participant observations of sign response and triggering from current software verification of evidence han-

dling. Historical repeated measures, held-out speech tests, and deployment records characterize component behavior.

A. Scripted Checks, Simulated Signs, and Natural Use

We recorded 42 sessions from 30 participants between 17 August and 5 September 2026. Session categories described the acquisition setting, not the event label. Six *scripted-action* sessions involved routine instructed B/E/F/A/S checks without target symptoms. Seven *simulated-positive* sessions included volunteers enacting facial droop, arm drift, trunk lean, or altered articulation. Five *challenge* sessions combined simulated signs with difficult inputs, including occlusion, low light, background sound, incomplete actions, and disconnection. Twelve natural-use sessions measured ordinary-activity triggering; 12 repeat sessions assessed repeated measurements. The 38 annotated targets comprised 28 events from simulated-positive sessions and 10 from challenge sessions.

The recorded configuration, v1.3.0+frozen (Full), used a personal Balance baseline and quality gates. Two historical comparison labels denoted a population Balance reference (Population baseline) or disabled gates (No gates). All three shared 201 event IDs. Annotations were recorded as blinded to system outputs. We analysed paired event outcomes with 10,000 participant-cluster bootstrap resamples (seed 20260910) and pointwise 95% intervals. The seed fixes resampling for reproducibility. Target response counted the recorded trigger label, without inferring passive detection or completed guided assessment. All 38 runtime paths remain unresolved; component names and acquisition timestamps do not establish their origin. Completeness indicated a completed measurement process linked to a report, independently of positivity.

Full had recorded triggers for 35/38 targets (92.1%) from 12 participants, versus 29/38 for Population baseline and 33/38 for No gates (Fig. 4). Face, Arms, Balance, and Speech counts were 9/10, 7/8, 10/11, and 9/9. The 15.8-percentage-point difference from Population baseline (95% interval 7.9–23.7) came entirely from six additional Balance detections. The difference from No gates was 5.3 points (0.0–14.7). Exact population-reference parameters and disabled rules remain unspecified, so these are descriptive contrasts between historical labels, not controlled ablations of the current implementation.

Among 177 started measurements from 25 participants, 159 were reference-complete and 18 were incomplete, including two aborted attempts. All configurations retained the 159 complete inputs. Incorrect completions were 1/18 for Full, 1/18 for Population baseline, and 15/18 for No gates. The Full-minus-No gates difference was -77.8 points (-95.0 to -55.6), describing a recorded completion difference. The unavailable gate configuration precludes attributing it to a specific quality rule.

During 29.164 reference-negative online hours from 12 natural-use participants, Full, Population baseline, and No gates recorded 14, 20, and 19 false triggers (0.480, 0.686, and 0.651 per hour). Online availability was 92.8%. These rates describe recorded triggers; complete capture coverage and linked

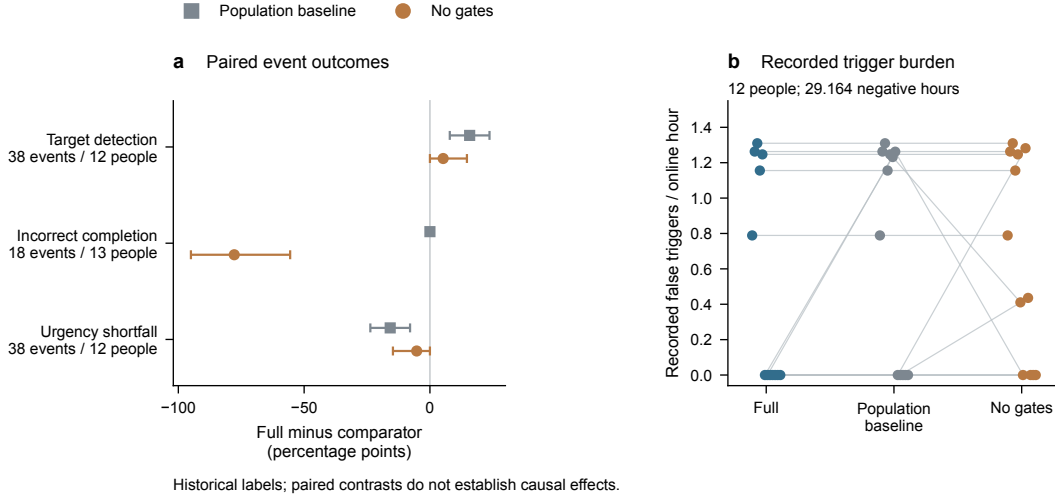


Fig. 4. Paired outcomes and recorded trigger burden. (a) Full-minus-comparator differences with 95% participant-bootstrap intervals; sample sizes appear beside each endpoint. (b) Participant rates over 29,164 reference-negative online hours. Lines connect the same participant; fixed horizontal offsets separate overlapping participants without altering rates. Rates describe the enumerated event set.

check outcomes are needed to quantify final-alert burden. Full also retained two symptom-count shortfalls among 35 known target records and one incomplete correction history among nine cases. Three target symptom counts were unknown. All three Full urgency shortfalls occurred among missed targets; detected targets had no urgency-label mismatches. Supporting notes retain extended endpoint and timing definitions.

Face reference values were recorded as the mean of two independent geometric annotations, with uncertainty listed as ± 0.006 but no statistical definition. The exported system values differed from these references by a mean absolute error (MAE) of 0.00452 interocular-width units (33 measurements, 22 participants; Fig. 5). The difference is comparable to the listed reference uncertainty and does not establish measurement accuracy. Eleven complete repeat pairs, separated by 11.2–15.6 days, yielded $\text{ICC}(2,1)=0.935$ (95% interval 0.801–0.975); one participant lacked the first measurement. Eyes exported gaze offset had MAE 0.280° against desktop eye-tracker values (25 measurements, 19 participants; system range 0.094–1.641 $^\circ$). Its runtime output mapping remains unresolved, so this comparison does not establish angular accuracy. Automatic Eyes abnormality classification remained disabled.

B. Historical Healthy-Participant Measurements

Forty healthy participants completed five visits each, yielding 200 sessions and 800 B/E/F/A records. Each module had 196 analyzable results, one positive (0.51%), and four insufficient outcomes (2.0% of attempts). Figure 6 reports $\text{ICC}(1,1)$ for one recorded output per module, with participant-bootstrap intervals (5,000 resamples; seed 203). B/E/F/A estimates were 0.140/0.266/0.784/0.679, indicating limited B/E repeatability relative to between-participant variation. The outputs were trunk-orientation change, binocular directional asymmetry, mouth-corner delta, and arm-level difference, respectively. These component outcomes do not estimate the

current workflow’s false-alert probability. Balance baseline linkage and Face shoulder-width normalization versus interocular normalization in the implementation require reconciliation. The source of the historical E positive label (questionnaire or automatic output) is unspecified; current automatic E classification is disabled. These statistics describe historical repeatability, not current threshold reliability.

C. Speaker-Independent Speech Evaluation

The Mandarin Dysarthria Speech Corpus (MDSC) contains 18,630 recordings from 46 Mandarin speakers: 21 with dysarthria and 25 controls [9]. Speaker-disjoint training, validation, and test sets contained 33/7/6 speakers and 13,365/2,835/2,430 recordings. The logistic model used Log-Mel, temporal, energy, and spectral descriptors. Its threshold was frozen at 0.735282 using validation sensitivity of at least 0.90 and maximal specificity. Intervals resampled speakers.

Table I reports held-out results. Sensitivity fell from 90.04% in validation to 67.24% in testing, while specificity rose from 81.36% to 98.85%. One speaker accounted for 313 of 398 misses (Fig. 7). Only six independent speakers contributed to testing, explaining the wide uncertainty despite 2,430 recordings. These results concern prompted dysarthric speech in quiet conditions; household and acute-stroke performance remain untested.

TABLE I
HELD-OUT SPEECH RESULTS: 2,430 RECORDINGS FROM SIX UNSEEN SPEAKERS. INTERVALS RESAMPLE SPEAKERS, NOT INDIVIDUAL RECORDINGS.

Metric	Estimate	95% interval
Sensitivity	67.24%	22.7–94.8%
Specificity	98.85%	98.0–99.8%
Balanced accuracy	83.05%	60.7–96.9%
ROC-AUC	0.9573	—
Brier score	0.0983	0.022–0.227

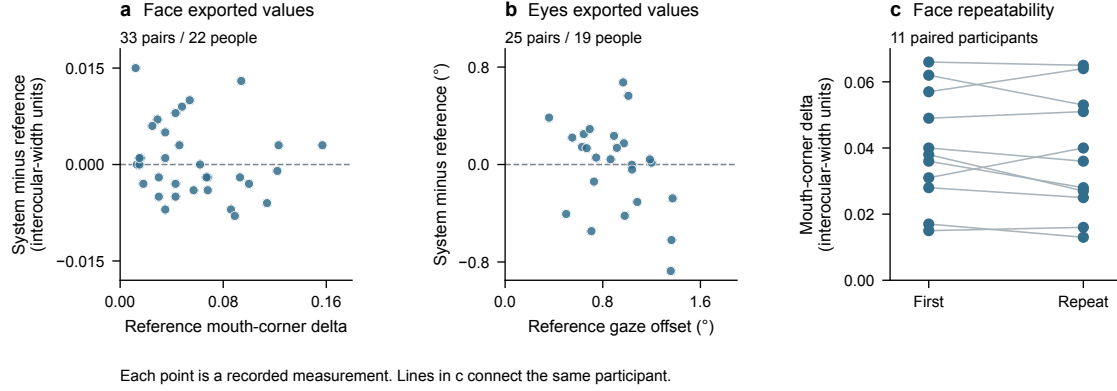


Fig. 5. Exported-value comparisons and Face repeatability. (a,b) System-minus-reference errors for Face mouth-corner change and Eyes gaze offset; dashed lines indicate zero difference. Eyes compares exported fields with unresolved runtime mapping. (c) Two system measurements for each of 11 participants with complete Face repeat pairs. Lines connect participants across visits.

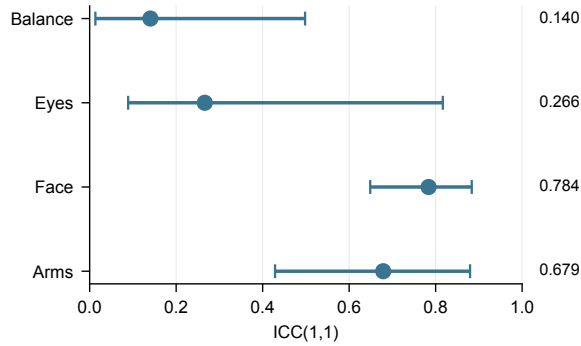


Fig. 6. Repeatability of one selected output per component, with 196 eligible measurements each. Bars show participant-bootstrap 95% intervals for ICC(1,1).

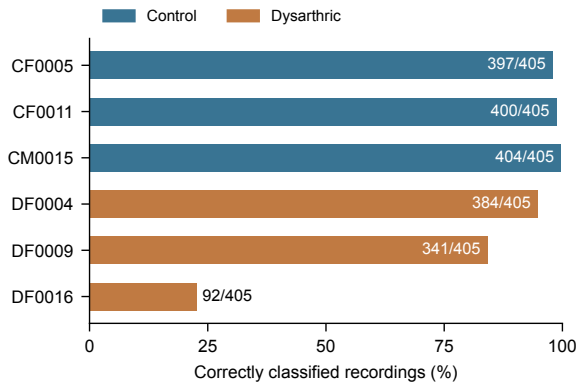


Fig. 7. Held-out speech performance at the frozen threshold, with 405 recordings per speaker. Control bars show specificity; dysarthric bars show sensitivity.

D. Pi-PC Deployment Pilot

Eight runs with two participants totaled 13 h, including one 8-h run, with 99 events and 780 resource samples. The setup linked a 4-GB Raspberry Pi 4, Camera V2, and ReSpeaker microphone HAT to an Intel NUC 13 Pro with a Logitech C920. The pilot used the repository implementation. Earlier

and later runs showed Balance target counts of 4/7 and 6/7, and Speech counts of 1/2 and 2/2. Negative-interval triggers were 2/11 and 1/11 for Balance, and 2/7 and 1/7 for Speech. Uncontrolled configuration changes and two participants limit this comparison to a descriptive observation. Supporting materials retain scenario counts.

During the 8-h run, CPU utilization averaged 43.1%, RAM 1500 MB, and temperature 69.4°C (maximum 83.4°C; Fig. 8). Records included one unavailability flag, two data-loss flags, and one restart. Sampling coverage therefore does not establish uninterrupted operation. Twenty logged *soon* outcomes remain unmapped to the current *urgent* rule. Speech timing lacks capture-window alignment, and the guided Balance record describes a questionnaire rather than 30-s standing acquisition. These records were not used to validate urgency agreement or automatic-check latency. External alert delivery was not configured.

E. Decision-Fusion and Interaction Verification

An independent append-only reference agreed with the implementation across 1,800 F/A/S transitions involving measurements, retries, symptoms, and corrections; previous report snapshots remained unchanged. In 15 constructed cases across B/E/F/A/S, removing symptom retention, snapshot isolation, or correction history broke the corresponding contract; the full implementation passed all cases. These constructed removals isolate failure modes.

Four API regression scenarios additionally passed, with 24 requests recorded: late audio after a symptom report, stale audio after reset, symptom correction and persistent history readback, and B/E retry with unknown onset. They exercise production API, session, reporting, and storage code using synthetic audio and stub recognition, without participant sensing. Traces and source hashes accompany the supporting materials.

The reducer matched all $5^5 = 3,125$ specified evidence combinations. Complete-first suppression differs in 2,070

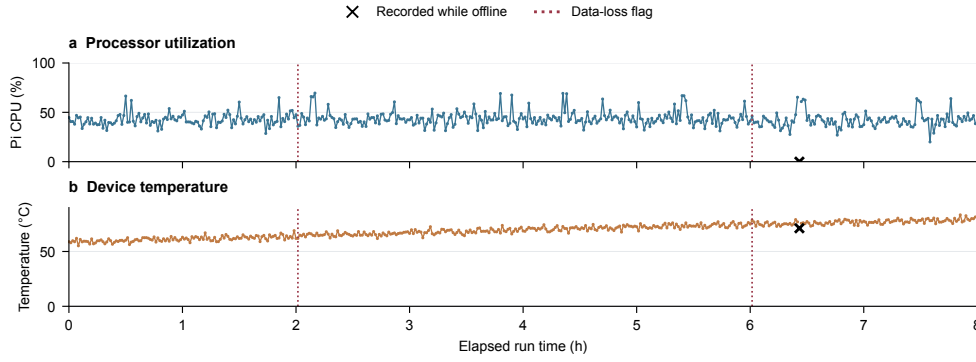


Fig. 8. CPU and temperature sampled every 60 s during the 8-h run. Crosses mark offline records; dotted lines mark data loss; traces break at offline records.

cases by construction, not measured benefit. Supporting materials retain the fixtures and protocols.

VII. DISCUSSION AND LIMITATIONS

The prototype links physical sensing and guided checks through evidence-handling contracts. State-transition and API tests verify symptom retention, correction history, and report isolation under constructed interactions. They establish implementation behavior; participant records separately describe target response, healthy measurements, and deployment.

These observations expose the remaining application question: can guided checks reject ordinary-activity triggers while retaining abnormal signs? The recorded 0.480 false triggers per reference-negative online hour represents a practical burden. Healthy-check positivity cannot establish a low final-alert rate because protocols, baselines, and output definitions differ. A frozen-configuration study linking every trigger, guided outcome, final alert, and missed target would test the workflow's benefit directly.

Historical configuration contrasts remain descriptive because comparator settings are unavailable. The complete-first rule comparison isolates a policy difference, while API traces verify synthetic interactions through reporting and storage. Neither reconstructs the historical symptom and history failures; their causes require original input sequences.

Component evidence also has specific limits: MDSC sensitivity varies across six test speakers; Face reference uncertainty is undefined; Eyes exported values lack a unique runtime mapping. Clinical sensitivity requires prospective clinical references, and predictive fusion requires same-session multimodal data with participant-level separation [14]. The two-person pilot supports physical deployment but records outages, data loss, and a restart. Continuous 24-h operation and external alert delivery remain untested.

VIII. CONCLUSION

PiBE-FAST integrates passive sensing, guided BE-FAST checks, and evidence-preserving interactions for stroke warning-sign assessment. Software verification and participant observations establish a working prototype and expose a remaining false-trigger burden. Reliable, low-burden warning requires linked final-alert outcomes and independent clinical validation.

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